Q1 JEE Main 2020 - 7 Jan (Morning)

Let $x^k + y^k = a^k$ where $a,\ k > 0$ and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$ then find k

(A) $\frac{1}{3}$
(B) $\frac{2}{3}$
(C) $\frac{4}{3}$
(D) 2

Q2 JEE Main 2020 - 7 Jan (Morning)

If $y(\alpha) = \sqrt{2\left(\frac{\tan\alpha + \cot\alpha}{1+\tan^2\alpha}\right) + \frac{1}{\sin^2\alpha}}, \alpha \in \left(\frac{3\pi}{4},\ \pi\right)$ then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is

(A) 4
(B) 2
(C) 3
(D) -4

Q3 JEE Main 2020 - 7 Jan (Evening)

Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to :

(A) $\frac{5}{\sqrt{7}}$
(B) $-\frac{\sqrt{5}}{4}$
(C) $-\frac{\sqrt{5}}{2}$
(D) $\frac{\sqrt{5}}{2}$

Q4 JEE Main 2020 - 9 Jan (Evening)

If $x = 2\sin\theta - \sin 2\theta$ and $y = 2\cos\theta - \cos 2\theta,\ \theta \in [0,\ 2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta = \pi$ is:

(A) $\frac{3}{8}$
(B) $\frac{3}{2}$
(C) $\frac{3}{4}$
(D) $-\frac{3}{4}$

Q5 JEE Main 2020 - 3 Sep (Morning)

If $y^2 + \log_e(\cos^2 x) = y, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then :

(A) $|y'(0)| + |y''(0)| = 3$
(B) $|y'(0)| + |y''(0)| = 1$
(C) $y''(0) = 0$
(D) $|y''(0)| = 2$

Q6 JEE Main 2020 - 4 Sep (Morning)

If $(a + \sqrt{2b}\cos x)(a - \sqrt{2}b\cos y) = a^2 - b^2$, where $a > b > 0$, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is :

(A) $\frac{2a+b}{2a-b}$
(B) $\frac{a-b}{a+b}$
(C) $\frac{a+b}{a-b}$
(D) $\frac{a-2b}{a+2b}$

Q7 JEE Main 2020 - 5 Sep (Evening)

The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at $x = \frac{1}{2}$ is :

(A) $\frac{2\sqrt{3}}{3}$
(B) $\frac{2\sqrt{3}}{5}$
(C) $\frac{\sqrt{3}}{10}$
(D) $\frac{\sqrt{3}}{12}$

Q8 JEE Main 2020 - 6 Sep (Morning)

Let $f : R \rightarrow R$ be defined as $f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x}\right) + 5x^2, x < 0 \\ 0, x = 0 \\ x^5 \cos\left(\frac{1}{x}\right) + \lambda x^2, x > 0 \end{cases}$ The value of λ for which $f''(0)$ exists, is

Answer Key

Q1 (B)	**Q2** (A)	**Q3** (C)	**Q4** (A)
Q5 (D)	**Q6** (C)	**Q7** (C)	**Q8** (5)

Q1

$$k \cdot x^{k-1} + k \cdot y^{k-1}\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^{k-1}$$

$$\frac{dy}{dx} + \left(\frac{x}{y}\right)^{k-1} = 0$$

$$k - 1 = -\frac{1}{3}$$

$$k = 1 - \frac{1}{3} = \frac{2}{3}$$

Q2

$$y = \sqrt{\frac{2\cos^2\alpha}{\sin\alpha\cos\alpha} + \frac{1}{\sin^2\alpha}} = \sqrt{2\cot\alpha + \operatorname{cosec}^2\alpha}$$

$$= |1 + \cot\alpha| = -1 - \cot\alpha$$

$$\frac{dy}{d\alpha} = \operatorname{cosec}^2\alpha \quad \Rightarrow \quad \left(\frac{dy}{dx}\right) \text{ at } \alpha = \frac{5\pi}{6} \text{ will be} = 4$$

Q3

$$x=\frac{1}{2},\ y=\frac{-1}{4}\quad\Rightarrow\quad xy=\frac{-1}{8}$$

$$y\cdot\frac{1\cdot(=2x)}{2\sqrt{-x^2}}+y'\cdot\sqrt{1-x^2}=-\left\{1\cdot\sqrt{1-y^2}+\frac{x\cdot(-2y)}{2\sqrt{1-y^2}}y'\right\}$$

$$-\frac{xy}{\sqrt{1-x^2}}+y'\sqrt{1-x^2}=-\sqrt{1-y^2}+\frac{xy\cdot y'}{\sqrt{1-y^2}}$$

$$y'\left(\sqrt{1-x^2}-\frac{xy}{\sqrt{1-y^2}}\right)=\frac{xy}{\sqrt{1-x^2}}-\sqrt{1-y^2}$$

$$y'\left(\frac{\sqrt{3}}{2}+\frac{1}{8\cdot\frac{\sqrt{15}}{4}}\right)=\frac{-1}{8\cdot\sqrt{\frac{3}{2}}}-\frac{\sqrt{15}}{4}$$

$$y'\left(\frac{\sqrt{45}+1}{2\sqrt{15}}\right)=-\frac{\left(1+\sqrt{45}\right)}{4\sqrt{3}}$$

$$y'=-\frac{\sqrt{5}}{2}$$

Q4

$$\frac{dx}{d\theta}=2\cos\theta-2\cos 2\theta$$

$$\frac{dy}{d\theta}=-2\sin\theta+2\sin 2\theta$$

$$\therefore\quad\frac{dy}{dx}=\frac{\sin 2\theta-\sin\theta}{\cos\theta-cso2\theta}$$

$$=\frac{2\sin\frac{\theta}{2}\cdot\cos\frac{3\theta}{2}}{2\sin\frac{\theta}{2}\sin\frac{3\theta}{2}}=\cot\frac{3\theta}{2}$$

$$\frac{d^2y}{dx^2}=\frac{-3}{2}\operatorname{cosec}^2\frac{3\theta}{2}\cdot\frac{d\theta}{dx}$$

$$\frac{d^2y}{dx^2}=\frac{-\frac{3}{2}\operatorname{cosec}^2\frac{3\theta}{2}}{2\left(\cos\theta-cos2\theta\right)}$$

$$\left.\frac{d^2y}{dx^2}\right|_{\theta=\pi}=-\frac{3}{4\left(-1-1\right)}=\frac{3}{8}$$

Q5

$y^2 + 2\ln(\cos x) = y$
$\Rightarrow 2yy' - 2\tan x = y'$
From (1) $\quad y(0) = 0$ or 1
$\therefore \quad y'(0) = 0$
Again differentiating
(2) we get
$2(y')^2 + 2yy'' - 2sec^2 x = y''$
gives $|y''(0)| = 2$

Q6

$(a + \sqrt{2}b\cos x)(a - \sqrt{2}b\cos y) = a^2 - b^2$
Differentiating both sides
$(-\sqrt{2}b\sin x)(a - \sqrt{2}\cos y) + (a + \sqrt{2}b\cos x)$
$(\sqrt{2}b\sin y)y' = 0$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$
$-b(a-b) + (a+b)by' = 0$
$\frac{dy}{dx} = \frac{a-b}{a+b}$
$\Rightarrow \frac{dx}{dy} = \frac{a+b}{a-b}$

Q7

$$\frac{d\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)}{d\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)} = \frac{\frac{d\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)}{dx}}{\frac{d\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)}{dx}}$$

Simplifying $\left(\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)\right)$ Put $x = \tan\theta$
$\Rightarrow \tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right) = \tan^{-1}\left(\frac{1-(1-2\sin^2\theta/2)}{2\sin\theta/2\cos\theta/2}\right) = \frac{\theta}{2}$
$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{\tan^{-1}x}{2}$
\& similarly $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ Put $x = \sin\theta$
$\Rightarrow \tan^{-1}\left(\frac{\sin 2\theta}{\cos 2\theta}\right) = 2\theta = 2\sin^{-1}x$

Hence required derivative

$$\frac{\frac{1}{2\left(1+x^2\right)}}{\frac{2}{\sqrt{1-x^2}}}=\left.\frac{\sqrt{1-x^2}}{4(1+x^2)}\right|_{x=\frac{1}{2}}=\frac{\sqrt{3}}{10}$$

Q8

If $g(x)=x^5\sin\left(\frac{1}{x}\right)$ and $h(x)=x^5\cos\left(\frac{1}{x}\right)$
then $g''(0)=0$ and $h''(0)=0$
So, $f''\left(0^+\right)=g''\left(0^+\right)+10=10$
and $f''\left(0^-\right)=h''\left(0^-\right)+2\lambda=f''\left(0^+\right)$
$\Rightarrow 2\lambda=10$
$\lambda=5$